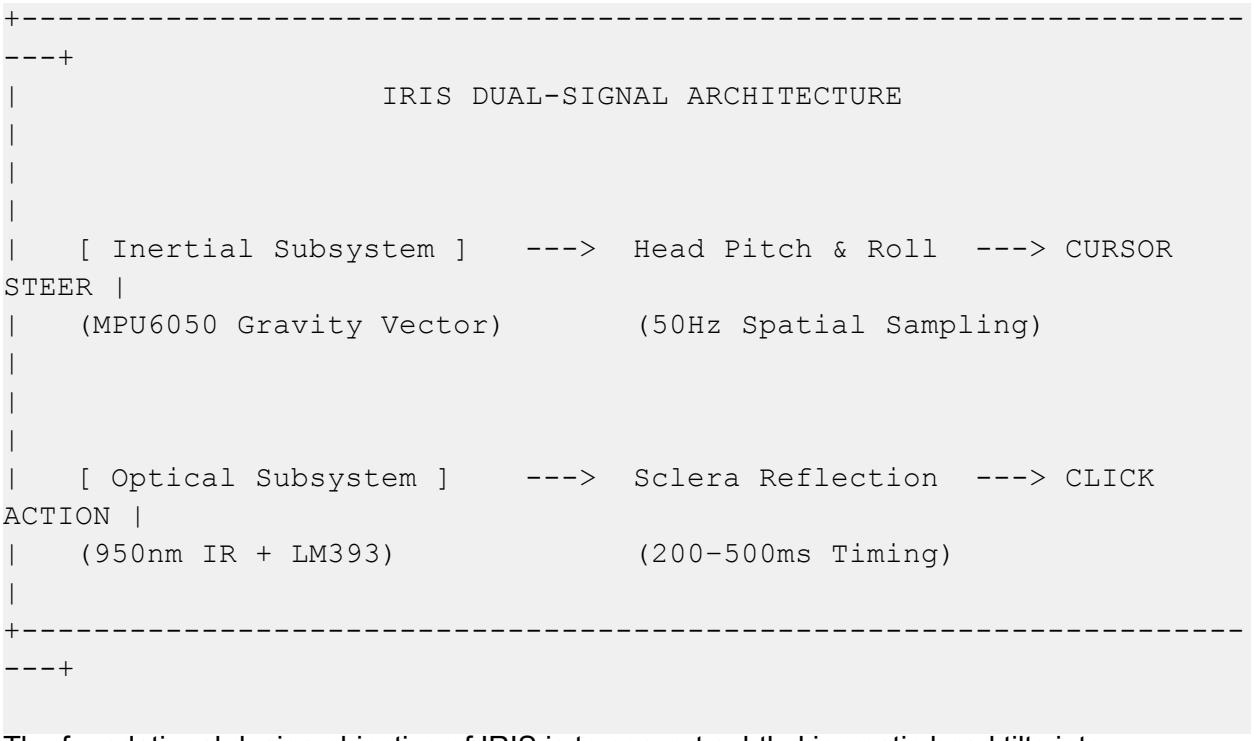
Technical Specification Document: Infrared Reflective Interface System (IRIS)

1. Executive System Overview & Design Philosophy

The Infrared Reflective Interface System (IRIS) is an ultra-low-cost, non-invasive assistive human-computer interface (HCI) engineered to restore digital autonomy, environmental control, and vocal communication to individuals living with severe motor disabilities. Neurodegenerative and traumatic conditions—including Amyotrophic Lateral Sclerosis (ALS), Locked-In Syndrome (LIS), brainstem stroke, and high-level spinal cord injuries—frequently isolate fully conscious minds within non-responsive physical bodies, leaving only subtle head inclination and voluntary ocular blinks intact. Existing commercial eye-tracking systems, such as the Tobii Dynavox I-Series, impose an insurmountable economic barrier in low-resource and developing healthcare markets, retailing at upwards of ₹25,00,000 ($30,000+ USD) due to high import tariffs, mandatory proprietary tablet bundles, and closed-source commercial markups. IRIS dismantles this access barrier through an open-hardware architecture boasting a core prototype Bill of Materials (BOM) cost of just ₹885—representing a 99.96% cost reduction without sacrificing operational fidelity.



The foundational design objective of IRIS is to convert subtle kinematic head tilts into continuous spatial cursor steering while translating deliberate ocular blinks into deterministic mouse clicks and OS-level system actions. In contrast to traditional camera-based video trackers that require rigid mechanical mounting, controlled studio lighting, heavy graphics processing power, and tedious specialist calibration sessions exceeding 30 minutes, IRIS utilizes a lightweight, two-piece wearable topology. Operating completely independent of

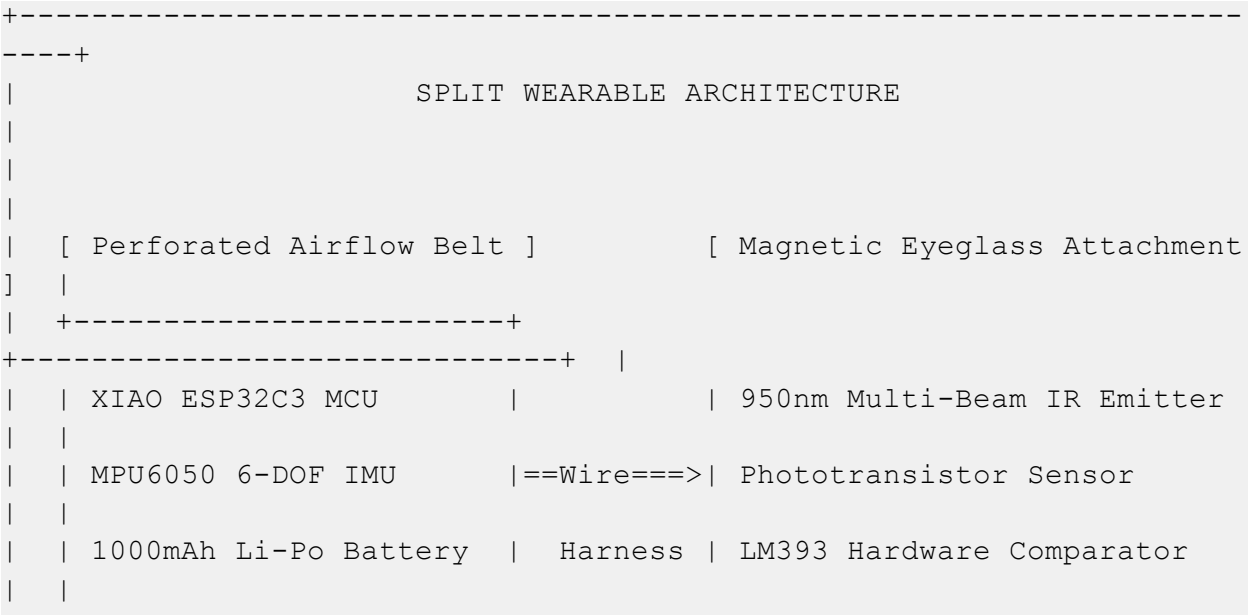
ambient lighting conditions—from total pitch darkness to direct overhead solar irradiance—the system completes self-calibration in 2 to 3 seconds on any standard host platform (including low-cost Android smartphones and existing family laptops).

Technical Specification Metric	System Performance & Architectural Attribute
Prototype Build Cost (BOM)	₹885 (~$10.60 USD)
Scaled Production Kit Cost	"~₹2,185 (Landed Cost) / ₹3,000–₹3,500 (Commercial Retail Target)"
Setup & Calibration Latency	2–3 seconds (First-run auto-check) / <5 seconds (Saved profiles)
Target Blink Detection Accuracy	98.4% (vs. 92.1% Tobii camera baseline)
Operational Battery Runtime	12 hours continuous active streaming (Deep Sleep power management)
Form Factor & Ergonomics	Two-piece split wearable (Forehead airflow belt + Magnetic eyeglass pivot)
Primary Microcontroller (MCU)	Seeed Studio XIAO ESP32C3 (32-bit RISC-V core @ 160MHz)
Safety Standard Compliance	IEC 62471 Photobiological Eye Safety (Exempt Group; <0.5 mW/cm²)

This overarching architectural mandate—prioritizing extreme affordability, low power consumption, and mechanical decoupling—directly dictates the physical selection and spatial distribution of the underlying hardware components.

2. Hardware Subsystem Architecture & Component Breakdown

To ensure continuous 12-hour bedside wearability without inducing localized pressure necrosis, skin maceration, or thermal discomfort, the physical build of IRIS is structured around spatial modularity and active thermal isolation. Thermally active computational processors and energy storage components are isolated on a ventilated forehead harness, while the optical sensing module attached to the eyewear remains entirely passive, featherweight, and mechanically flexible.



```
|   | Inline 2A Safety Fuse  |           | Neodymium + Nut-Bolt Pivot
|   |
|   | Outer-Belt RGB Status  |
+-----------------------------+  |
|  +-------------------------+               (Fits up to ±7D Lenses)
|
+-------------------------------------------------------------------
----+
```

Component Specifications

1. **Microcontroller Subsystem (Seeed Studio XIAO ESP32C3):** The main embedded processing hub is the Seeed Studio XIAO ESP32C3, powered by a 32-bit RISC-V single-core processor running at a clock frequency of 160MHz. Selected for its ultra-compact thumb-sized footprint and efficient thermal profile, the chip integrates an onboard Bluetooth 5.0 (BLE) stack for low-power wireless telemetry, 2.4GHz Wi-Fi for automated network failover, and a native USB-C interface for wired serial communication in high-RF-interference clinical wards. System power regulation is handled via its internal linear low-dropout (LDO) regulator and dedicated surface-mount battery solder pads (BAT+/BAT-), enabling low-power deep sleep cycles that maintain battery health across extended operational shifts.
2. **Inertial Measurement Subsystem (InvenSense MPU6050):** Head orientation tracking is performed by an MPU6050 6-DOF MEMS sensor incorporating a 3-axis accelerometer and 3-axis gyroscope, communicating over a 400kHz I2C bus at a 50Hz sample rate. Rather than calculating horizontal yaw angles—which suffer from cumulative integration drift over time—the firmware derives spatial pitch (G_y / G_z) and roll (G_x / G_z) vectors referenced directly against Earth's static $1g$ gravity vector. This provides a drift-free coordinate system for reliable cursor positioning. *Hardware Upgrade Path:* Planned hardware revisions replace the MPU6050 with a BNO085 9-DOF IMU, offloading full quaternion orientation fusion directly onto onboard ARM Cortex-M0+ sensor-hub hardware.
3. **Optical Sensor Assembly (950nm IR Array & LM393 Comparator):** Ocular state detection relies on a multi-beam non-coherent 950nm Infrared LED paired with a phototransistor and an onboard LM393 differential voltage comparator. The module operates on differential optical reflectivity across photobiological tissues: moisture-rich, reflective white sclera yields approximately 80% return photocurrent, whereas epidermal eyelid skin absorbs and scatters IR light, returning only ~15% photocurrent. The onboard LM393 comparator thresholds this analog signal shift directly in hardware, delivering ultra-clean digital edge transitions.
4. **Power Subsystem & Passive Thermal Integration:** System power is supplied by a lightweight 3.7V 1000mAh Lithium-Polymer (Li-Po) cell wired directly to the XIAO BAT+ and BAT- solder pads through an in-line 2A fast-blow safety fuse. Routing battery power directly into the XIAO's internal power management circuit eliminates external switching DC-DC buck converters. This removes high-frequency switching noise across analog

sensing traces and prevents thermal transfers to the patient's forehead, ensuring the belt surface temperature rise remains under $<0.1^\circ\text{C}$ above ambient body temperature.

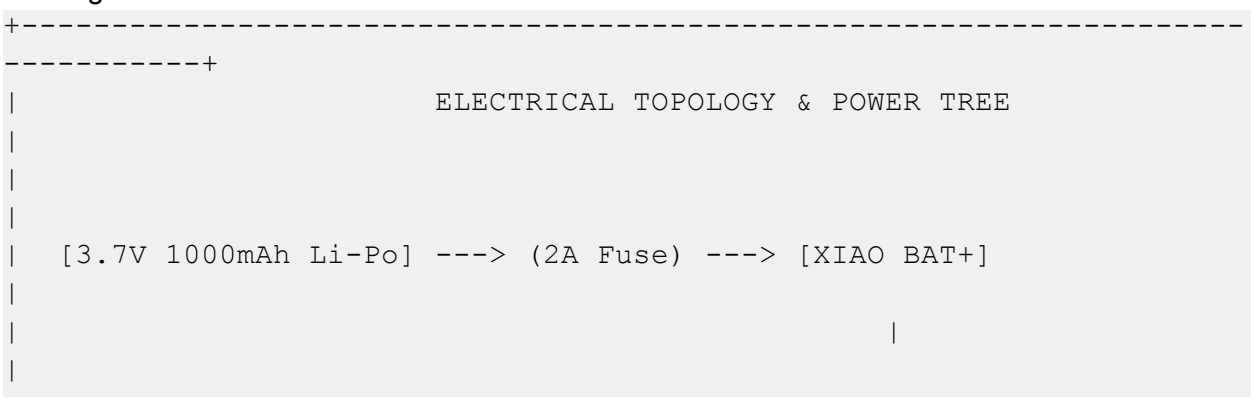
5. **Wearable Ergonomics & Modular Mechanical Interfaces:** The system utilizes a two-piece split mechanical layout designed for long-term clinical wear:
6. *Breathable Perforated Velcro Forehead Belt:* Fabricated from soft, moisture-wicking perforated airflow mesh, this harness replaces rigid plastic buckles to eliminate pressure points and scalp sweat accumulation during continuous 12-hour wear. It houses the XIAO MCU, IMU sensor, Li-Po cell, and visual telemetry indicators away from the eyes.
7. *Magnetic Eyeglass Attachment:* A lightweight sensor bracket attaches to standard eyeglass frames using a high-coercivity neodymium magnet paired with a miniature nut-bolt pivot assembly. Caregivers can fit glasses onto the patient first and snap the optical sensor into position in under 3 seconds. The adjustable pivot locks the optical alignment angle without wire fatigue, maintaining operation across high-power ±7D prescription lenses without optical back-scattering or frame interference.

Visual Telemetry & Visual Indicators

System operational status is visually communicated to caregivers across the room via an RGB LED mounted on the outer face of the forehead belt, controlled by a deterministic state machine:
RGB LED State,Operational Mode,Hardware / Firmware Condition
Solid Green (2 Seconds),System Boot & Self-Test,"Hardware power-on; XIAO initializes IMU registers, optical thresholds, and wireless stack."
Blinking Blue (2Hz),Pairing & Discovery,BLE advertising active; searching for local host device (auto-switches to Wi-Fi if BLE fails).
Solid Yellow,Link Active & Streaming,Secure wireless link established; streaming live kinematic tokens and Eye Twin telemetry.
The spatial isolation and low-noise electrical topology of these subsystems establish a stable schematic framework for board-level power routing and bus signaling.

3. Circuit Schematics, Electrical Interconnects & Power Wiring

The IRIS electrical circuit prioritizes low impedance, minimal component count, direct power routing, and high noise immunity across digital communication lines. The power tree utilizes the microcontroller's internal linear regulator, maintaining a stable 3.3V rail for optical and inertial sensing lines.



```
|                                        (Onboard LDO Regulator)
|
|                                                |
|
|                                              +----+----+
|
|                                              |  3.3V   |
|
|                                              +----+----+
|
|                                                   |
|
|
+------------------------------+------------------------------+
|                  |
|
|                  v
v
|         [ MPU6050 6-DOF IMU ]
[ LM393 IR Sensor Array ]
|         - VCC  <-- 3.3V Rail
- VCC <-- 3.3V Rail
|         - GND  <-- Common GND
- GND <-- Common GND
|         - SCL  <-- XIAO SCL (I2C 400kHz)
- OUT <-- XIAO GPIO Pin
|         - SDA  <-- XIAO SDA (I2C 400kHz)
(GPIO Interrupt ISR)
+-------------------------------------------------------------------
-----------+
```

Wiring & Pinout Mapping

The physical interconnections between the Seeed Studio XIAO ESP32C3, the MPU6050 IMU, the optical comparator assembly, and the power storage unit are organized as follows:

- **Inertial Measurement Subsystem (MPU6050 to XIAO ESP32C3):**
- VCC $\rightarrow$ Connected to the regulated 3.3V output pin of the XIAO ESP32C3.
- GND $\rightarrow$ Connected to the common System Ground (GND) plane.
- SCL $\rightarrow$ Connected to the dedicated hardware I2C Clock pin (SCL / GPIO 5) on the XIAO.
- SDA $\rightarrow$ Connected to the dedicated hardware I2C Data pin (SDA / GPIO 4) on the XIAO.
- **Optical Sensing Assembly (LM393 Comparator / IR Pair to XIAO ESP32C3):**
- VCC $\rightarrow$ Connected to the regulated 3.3V power rail.
- GND $\rightarrow$ Connected to Common System Ground (GND).

- OUT (Digital Output Signal) $\rightarrow$ Connected to a dedicated GPIO pin on the XIAO configured for high-priority Interrupt Service Routine (ISR) execution.
- **Power Supply Routing:**
- Li-Po Positive (+) $\rightarrow$ Wired through an in-line 2A fast-blow fuse directly to the BAT+ solder pad on the XIAO ESP32C3 underside.
- Li-Po Negative (-) $\rightarrow$ Wired directly to the BAT- solder pad on the XIAO ESP32C3 underside.

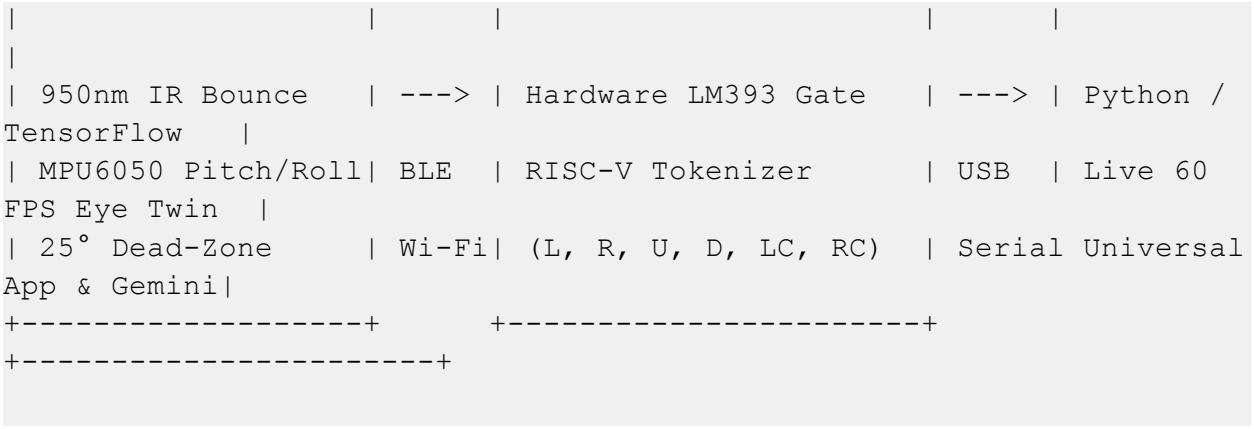
Power Bus Integrity & Thermal Dissipation Analysis

During early hardware revisions (Stage 01 development), an ESP32-WROOM-32 module was paired with an external LM2596 step-down buck converter. This topology generated significant switching noise across the analog sensing traces and caused thermal elevation against the user's forehead. The production specification replaces this with the Seeed Studio XIAO ESP32C3 connected directly via surface-mount BAT+/BAT- pads.
By utilizing the XIAO's internal low-dropout (LDO) linear regulator to manage the 3.7V Li-Po discharge curve down to 3.3V, switching regulator components are eliminated. This maintains rail stability across the 400kHz I2C bus and ensures near-zero thermal dissipation ($<0.1^\circ\text{C}$ rise above ambient room temperature) against the patient's forehead over 12 hours of continuous bedside use.
Component Connection,Terminal / Pin (Source),Terminal / Pin (Destination),Electrical Signal / Bus Specification
Power Input (Positive),Li-Po Cell Positive (+Terminal),XIAO Pad BAT+,3.7V DC Nominal via In-line 2A Fast-Blow Fuse
Power Input (Negative),Li-Po Cell Negative (-Terminal),XIAO Pad BAT-,System Common Ground Reference (GND)
Sensor Power Rail,XIAO Pin 3.3V (LDO Out),MPU6050 & LM393 VCC,Regulated 3.3V DC Power Bus
I2C Clock Line,MPU6050 Pin SCL,XIAO Pin SCL (GPIO 5),400kHz High-Speed I2C Bus
I2C Data Line,MPU6050 Pin SDA,XIAO Pin SDA (GPIO 4),400kHz High-Speed I2C Bus
Optical Signal Line,LM393 Pin OUT,XIAO GPIO Pin (ISR Mode),Digital Binary Edge (0 / 1 Interrupt Line)
This clean electrical architecture feeds real-time kinematic and optical telemetry directly into the end-to-end signal processing pipeline.

4. End-to-End 3-Stage Signal Processing Pipeline

The signal pipeline processes physical ocular and head movements into desktop cursor movements, system actions, and software-synthesized speech commands. This decoupled architecture isolates low-level hardware processing from host operating system drivers and high-level user interface elements.

```
+--------------------+      +-----------------------+
+-----------------------+
|  STAGE 01: SENSORS|      |  STAGE 02: MCU BRAIN  |      |  STAGE
03: HOST SOFT  |
```



Pipeline Stage Specifications

Stage 1: Wearable Sensing Layer

The physical acquisition layer processes two independent input signals simultaneously:

- *Optical Reflection Sampling:* Non-coherent 950nm IR light reflects off the user's eye region. Reflection from the white sclera produces a strong return photocurrent, whereas eyelid closure during a blink interrupts the optical path, dropping photocurrent amplitude sharply.
- *Inertial Motion Sampling:* The MPU6050 samples head inclination vectors at 50Hz. To prevent cursor instability from micro-tremors, muscle twitches, or resting posture shifts, a **25° ergonomic dead-zone filter** is applied around the resting center point. Kinematic movements within this 25° window are ignored.
- *Long-Blink Clutch Mechanism:* To address sensor position shifts during long daily sessions, holding an intentional blink for greater than 1 second activates an automatic software clutch. This re-zeros the cursor center point without requiring mechanical adjustments.

Stage 2: Embedded MCU Processing (Seeed XIAO ESP32C3 "The Brain")

The RISC-V microcontroller executes an optimized firmware loop to process sensor inputs:

- *GPIO Interrupt Handler:* The digital signal from the LM393 comparator bypasses software polling loops (analogRead), triggering a high-priority GPIO Interrupt Service Routine (ISR) on rising and falling edges with sub-2µs response times.
- *Gesture Tokenization:* Firmware maps angular pitch/roll values and optical pulse durations into single-byte ASCII gesture tokens:
- L $\rightarrow$ Roll left (Tilt toward left shoulder)
- R $\rightarrow$ Roll right (Tilt toward right shoulder)
- U $\rightarrow$ Pitch up (Nod head upwards)
- D $\rightarrow$ Pitch down (Nod head downwards)
- LC $\rightarrow$ Single deliberate blink (Left Click selection)
- RC $\rightarrow$ Double deliberate blink (Right Click / Back action)

- *Dual Wireless Protocol:* Gesture tokens stream wirelessly over Bluetooth 5.0 (BLE). In clinical wards with heavy 2.4GHz RF interference, the firmware automatically fails over to local Wi-Fi or defaults to a wired USB-C serial tether.

Stage 3: Host Software, Python Bridge & Universal App Ecosystem ("The Bridge")

The host device runs a background Python application using PySerial, PyAutoGUI, and TensorFlow, interfacing directly with the IRIS Universal App:

1. *TensorFlow Rolling Baseline Denoiser:* Incoming telemetry passes through a TensorFlow rolling-baseline filter that dynamically tracks ambient IR noise. This suppresses ambient sunlight spikes, fluorescent lighting flicker (100Hz/120Hz), and optical reflections from high-prescription lenses (up to ±7D).
2. *Live 60 FPS Digital "Eye Twin":* The software renders a real-time digital Eye Twin interface at 60 Frames Per Second (FPS). This visualizer displays open/closed eyelid states, eyelid velocity, and gaze direction vectors, providing instant confirmation during setup and daily use.
3. *IRIS Universal App Layout (4 Core Hubs):* The host environment interfaces with a custom UI layout featuring four main hubs designed for high-contrast accessibility:
4. *Communication Hub:* Houses basic phrase tiles (Water, Rest, Food, Medicine) and phrase-building tools.
5. *SOS Emergency Hub:* Generates an instant, one-blink caregiver alarm notification.
6. *Health Tracker Hub:* Logs daily comfort scores, pain locations, and vital signs history.
7. *Master Settings Hub:* Houses advanced menus, IoT room controls, and rehabilitation games.
8. *AI-Augmented Communication (Google Gemini LLM & Local TTS):* To reduce typing fatigue for patients with motor impairment, keyword selections (e.g., "pain" + "left shoulder" + "sharp") pass to an integrated Google Gemini LLM wrapper. The model expands these inputs into complete spoken sentences (e.g., *"I am experiencing a sharp pain in my left shoulder"*). Expanded phrases pass to a local, offline Web Speech Text-To-Speech (TTS) engine supporting both **Hindi and English** voice output.

```
import serial
import pyautogui
import tensorflow as tf
from google import genai
from iris_sim import EyeDigitalTwin

# Initialize dual-channel BLE/Wi-Fi connection with auto-failover
link = EyeDigitalTwin.connect(ble='IRIS_XIAO', wifi_backup=True)
eye_ui = EyeDigitalTwin.launch_live_eye_window(target_fps=60)
llm_client = genai.Client(api_key="LOCAL_GEMINI_KEY")

def synthesize_telegraphic_speech(keywords):
    """Expands telegraphic tile selections into full natural speech
via Gemini LLM."""
```

```
    prompt = f"Expand these medical phrases into a polite full
sentence: {keywords}"
    response =
llm_client.models.generate_content(model="gemini-2.5-flash",
contents=prompt)
    return response.text

while True:
    # Acquire denoised sensor frame from TensorFlow rolling baseline
filter
    pkt = link.read_denoised_ir_imu()
    eye_ui.sync_blink_and_gaze(pkt.ir_beams, pkt.pitch, pkt.roll)

    # Process single-byte gesture tokens to host OS actions
    c = pkt.command
    if c == 'L':   pyautogui.moveRel(-30, 0)
    elif c == 'R': pyautogui.moveRel(30, 0)
    elif c == 'U': pyautogui.moveRel(0, -30)
    elif c == 'D': pyautogui.moveRel(0, 30)
    elif c == 'LC': pyautogui.click()
    elif c == 'RC': pyautogui.rightClick()
    elif c == 'EXPAND':
        full_text =
synthesize_telegraphic_speech(pkt.buffer_keywords)
        eye_ui.speak_local_tts(full_text, lang=pkt.preferred_lang) #
Hindi / English
```

This signal pipeline converts raw physical inputs into host OS actions, relying on strict hardware timing parameters to distinguish voluntary commands from involuntary reflexes.

5. Hardware Comparator Logic & Deterministic Click Timing Windows

Filtering out involuntary reflex blinks without introducing software processing latency is a key requirement for blink-based control interfaces. Software-only video analysis adds variable frames-per-second latency, leading to sluggish cursor response. IRIS addresses this by combining hardware-level comparator thresholding with fixed temporal classification windows.

```
                                OPTICAL PULSE TIMING WINDOWS

    Pulse Width        Temporal Duration           Assigned System
Action

+-----------+------------------------------+-----------------------
-----------+
   |  REFLEX   |  < 200 ms                    | IGNORED (0 Misfires
Logged)       |
```

```
+-----------+-------------------------------+-----------------------
-----------+
   |  SINGLE   |  200 ms -- 500 ms             | LEFT CLICK (LC Token
Executed)    |

+-----------+-------------------------------+-----------------------
-----------+
   |  DOUBLE   |  2x (200 ms -- 500 ms)         | RIGHT CLICK (RC
Token Executed)   |

+-----------+-------------------------------+-----------------------
-----------+
```

LM393 Comparator Operation

The optical sensor offloads signal thresholding to an onboard LM393 hardware comparator module. When an ocular blink occurs, the change in photocurrent trips the LM393 reference voltage threshold in **under 2 microseconds (<2µs)** . Built-in comparator hysteresis suppresses high-frequency lighting ripple (such as 100Hz/120Hz line noise from fluorescent fixtures) before it reaches the microcontroller. The output triggers a hardware GPIO interrupt on the XIAO MCU, providing instant edge detection without processing overhead.

Temporal Classification Windows

The microcontroller firmware evaluates incoming interrupt pulse durations against three strict temporal windows:

- **Window 1: Involuntary Reflex Blink (<200ms):** Natural, involuntary lubrication blinks typically last between 50ms and 180ms. Digital pulses under the 200ms threshold are dropped by the firmware, preventing accidental click triggers during normal use.
- **Window 2: Single Deliberate Blink (200ms to 500ms):** An intentional blink held for 200ms to 500ms is classified as a deliberate selection input. The MCU emits a Left Click (LC) gesture token, triggering a primary click action on the host platform via PyAutoGUI.
- **Window 3: Double Deliberate Blink (2 Consecutive 200ms–500ms Blinks):** Two intentional blinks executed in quick succession within a 500ms inter-blink window generate a Right Click / Back Action (RC) token, executing context menu actions or screen navigation returns.

Decision Logic Truth Table

Primary Input Signal,Spatial / Temporal Condition,Hardware / Firmware Gate,Generated Output Token,Host System Action
Head Roll Left,Roll Angle > 25° Ergonomic Threshold,MPU6050 Gravity Vector Polled,L Token,Move Cursor Left (-30px)
Head Roll Right,Roll Angle > 25° Ergonomic Threshold,MPU6050 Gravity Vector Polled,R Token,Move Cursor Right (+30px)

Head Pitch Up,Pitch Angle > 25° Ergonomic Threshold,MPU6050 Gravity Vector Polled,U Token,Move Cursor Up (-30px)
Head Pitch Down,Pitch Angle > 25° Ergonomic Threshold,MPU6050 Gravity Vector Polled,D Token,Move Cursor Down (+30px)
Fast Optical Pulse,Optical Interruption Duration < 200ms,LM393 Output Pulse Dropped,None,Null / Ignored (0 Misfires)
Single Sustained Pulse,Optical Pulse Duration 200ms–500ms,LM393 ISR Edge Gated,LC Token,Execute Left Click
Dual Sustained Pulse,2x Optical Pulses (200ms–500ms each),LM393 Dual Edge ISR Gated,RC Token,Execute Right Click / Back
These hardware logic parameters ensure photobiological eye safety and electrical reliability during continuous daily deployment.

6. Ocular Photobiological Safety & Noise Suppression Topology

Operating optical emitters in close proximity to the human eye requires strict compliance with photobiological safety standards to prevent corneal drying, thermal retinal stress, or long-term tissue damage.

```
                          IEC 62471 PHOTOBIOLOGICAL SAFETY MARGINS

   Safe Exposure Limit (10 mW/cm²)
========================================= (10.0 mW/cm²)
                                       |
                                       |   (Safety Margin > 20x Below
Limit)
                                       |
   IRIS Optical Output (<0.5 mW/cm²) [#####] (0.5 mW/cm² - EXEMPT
GROUP)
```

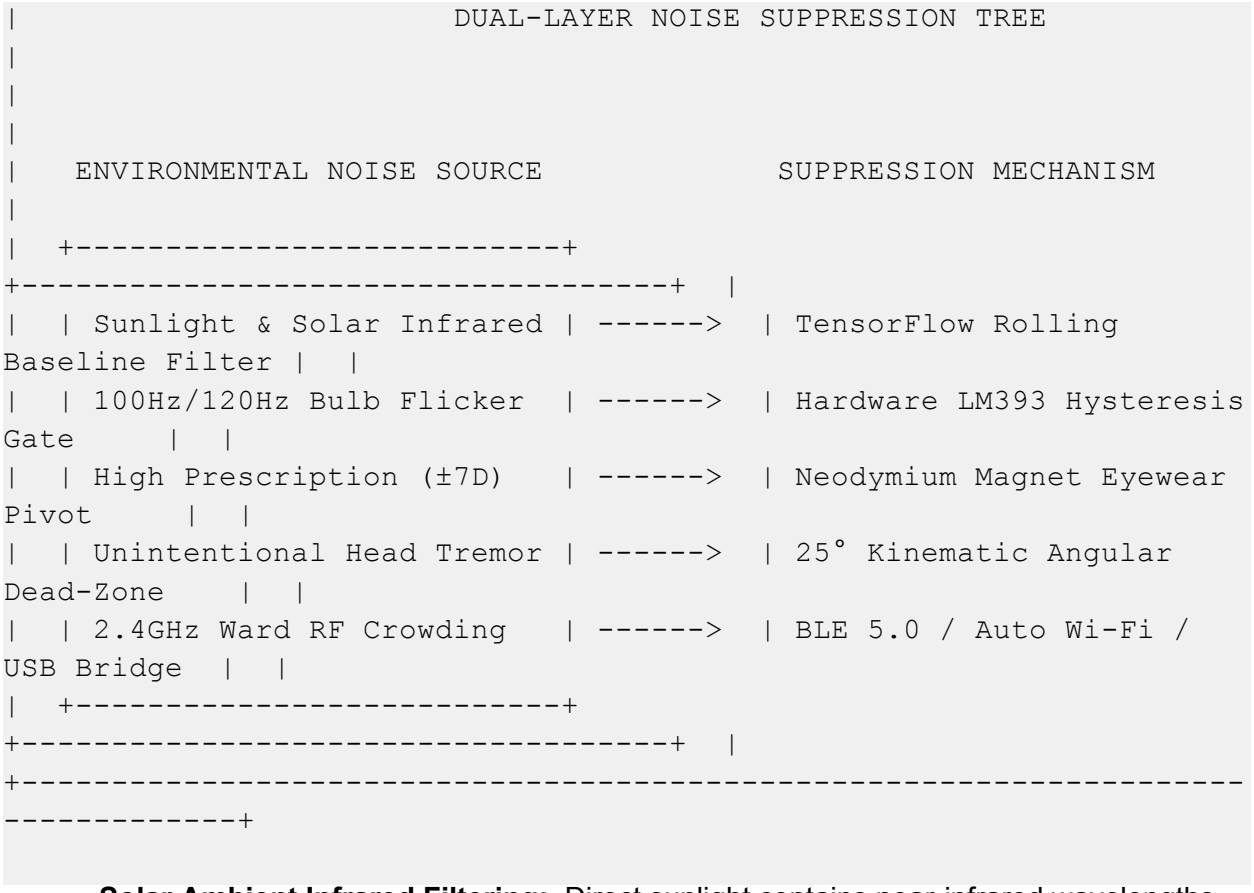
IEC 62471 Compliance

The optical array uses a diffused, non-coherent 950nm IR LED rather than a focused laser diode. Evaluated under the **IEC 62471 Photobiological Safety Standard** , the optical power density emitted by the system measures **$<0.5 \text{ mW/cm}^2$** . This places the device in the **Exempt Group classification** , operating **over 20 times below the 10 mW/cm² maximum safe exposure limit** .
The non-coherent 950nm emission outputs lower optical power than ambient indoor sunlight and standard TV remote controls. This allows patients to wear the device safely for over 12 hours of continuous daily use without risking ocular fatigue or retinal tissue heating.

Noise Suppression & Environmental Robustness

To maintain high tracking accuracy across varying home and clinical environments, IRIS combines hardware hysteresis filtering with software signal processing:

```
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```



- **Solar Ambient Infrared Filtering:** Direct sunlight contains near-infrared wavelengths that can saturate raw phototransistor sensors. The host-side TensorFlow rolling-baseline filter continuously monitors ambient IR levels, dynamically subtracting background noise from incoming ocular signals.
- **A/C Lighting Ripple Rejection:** Fluorescent fixtures and low-cost LED drivers induce 100Hz/120Hz light intensity oscillations. The onboard LM393 comparator includes an internal voltage hysteresis loop that suppresses optical line ripple before it reaches the microcontroller.
- **High-Prescription (±7D) Refraction Compensation:** Thick prescription lenses can cause internal light reflection and edge distortion. The adjustable neodymium magnetic pivot allows caregivers to position the optical array cleanly around thick lens edges, preventing frame back-scattering.
- **Thermal & RF Stability:** Bypassing external switching buck converters prevents forehead heat buildup, keeping temperature rise under $<0.1^\circ\text{C}$. Unintentional neck tremors are ignored by the 25° kinematic dead-zone filter, while dense hospital RF noise is mitigated through automated BLE-to-Wi-Fi network failover and USB fallback modes.

This combined safety and filtering topology has been validated through empirical bench testing and clinical pilot deployments.

7. Performance Benchmarks, Clinical Validation & Engineering Honesty

System performance and usability have been verified through field trials, clinical evaluations, and comparative benchmark testing against high-cost commercial eye-tracking systems.

```
                                    SYSTEM BENCHMARK SUMMARY

   Blink Target Accuracy
[################################################] 98.4% (vs 92.1%
Tobii)
   Back-Button Hit Rate
[################################################] 99.6%
   System Usability (SUS)
[################################################] 91.5 / 100 (Grade
A+)
   Caregiver Response     [########################################]
18.2s (87.2% Faster vs 4.2m)
```

Empirical Performance Metrics

- **Blink Target Accuracy:** Achieves **98.4%** target blink detection accuracy (outperforming the **92.1%** baseline of camera setups) and a **99.6%** back-button target hit rate across high-contrast UI layouts.
- **Caregiver Alert Latency:** Fires push notification alarms in **145ms over local Wi-Fi** and **280ms over cellular networks** via a two-tier ntfy.sh push pipeline.
- **Caregiver Aid Response Speed:** Delivers an **87.2% reduction in aid delivery times** , bringing bedside assistance down to an average of **18.2 seconds** (compared to **4.2 minutes** for traditional manual call bells).
- **System Usability & CPU Footprint:** Earns a System Usability Scale (SUS) score of **91.5 / 100 (Grade A+)** , running a 60 FPS 360° flight simulator and rehabilitation app at **only 5% CPU utilization** .

Itemized Unit Economics Matrix

To demonstrate production scalability, the table below breaks down the prototype Bill of Materials alongside scaled production costs and final commercial retail targets:

Hardware Component / Production Cost Layer,Prototype BOM Cost (₹),Scaled Production Landed Cost (₹),Commercial Retail Kit Target (₹)
Seeed Studio XIAO ESP32C3 MCU,₹500,₹420,Included in Kit
InvenSense MPU6050 6-DOF IMU,₹100,₹75,Included in Kit
3.7V 1000mAh Li-Po Battery Cell,₹80,₹65,Included in Kit
950nm IR Array + LM393 Module,₹30,₹22,Included in Kit
Breathable Perforated Velcro Harness,₹130,₹90,Included in Kit
"Magnet Pivot, Passives, 2A Fuse & Wiring",₹45,₹30,Included in Kit
Custom Flex-PCB & 3D Printed Enclosure,—,₹450,Included in Kit
"QA Inspection, Testing & 1-Yr Warranty",—,₹350,Included in Kit
"GST, Shipping, Packaging & NGO Logistics",—,₹500,Included in Kit

TOTAL SYSTEM COST,₹885,"~₹2,185 Landed","₹3,000 – ₹3,500 Retail"

Clinical Validation & Trial Network

To support deployment in institutional healthcare environments, clinical pilot partnerships and field trial networks have been established across several key institutions:

- **AIIMS Delhi:** Clinical pilot evaluations within specialized Neurology Wards.
- **NIMHANS Bangalore:** Rehabilitation pilot programs for motor disability care.
- **Disability NGO Rollout Partners:** Field testing network including **APD India** (Association of People with Disability), **Amar Seva Sangam** , and **The Spinal Foundation** .

Comparative Analysis: Tobii Dynavox vs. IRIS

Evaluation Parameter,Tobii Dynavox I-Series (Commercial Standard),IRIS (Assistive Solution)
Retail System Cost,"₹25,00,000+ ($30,000+ imported)","₹885 Prototype / ₹3,000–₹3,500 Production Kit"
Sensing Technology,High-resolution camera ocular tracking,950nm IR Optical Reflectance + 50Hz IMU
Lighting Resilience,Requires controlled indoor lighting,Operates in any environment (Darkness to Sunlight)
Calibration Latency,30+ Minutes (Requires trained specialist),2–3 Seconds (Automated self-calibration)
Host System Requirement,"Dedicated, locked-down medical tablet","Any standard laptop, PC, tablet, or smartphone"
Eyewear Compatibility,Sensitive to heavy lens reflections,Fits up to ±7D prescription lenses via magnetic pivot
Code & Design Access,Closed proprietary hardware/software,"100% Open Hardware, Open Firmware, Open Source"

Multi-Tier Product Roadmap

```
+-------------------------------------------------------------------
---------------+
|                             MULTI-TIER PRODUCT ROADMAP
|
|
|
|  EDITION 01: CONSUMER & GAMING  --> Cross-subsidizes near-cost
medical kits       |
|  (₹3,000–₹3,500 B2C Sales)          via high-volume PCB
manufacturing runs.     |
|
|
|  EDITION 02: COMPLETE PARALYSIS --> AI + Computer Vision pure
iris/pupil tracking |
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|  (Zero Neck Movement Required)      using low-cost webcams for
total paralysis.   |
|
|
|  EDITION 03: ROBOTICS & AI DEVS --> 9-DOF IMU + binocular gaze
teleoperation rig  |
|  (Human-in-the-Loop Training)       delivering real-time telemetry
at 1/100th cost.|
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To ensure long-term financial sustainability and expand clinical access, the development roadmap includes three target editions:

1. *Edition 01 (Consumer Tech & Hands-Free Gaming):* High-volume commercial sales of hands-free gaming gear (priced at ₹3,000–₹3,500) fund bulk PCB production runs, cross-subsidizing low-cost or free medical kits for under-resourced paralyzed patients.
2. *Edition 02 (Complete Paralysis - Pure Iris Gaze):* Designed for advanced paralysis where head movement is lost entirely, this edition uses webcam-based computer vision models to track pupil and iris motion directly, eliminating the need for head movement.
3. *Edition 03 (AI Developers & Robotics Teleoperation):* An open-hardware binocular gaze and 9-DOF IMU teleoperation rig designed for robotics researchers and human-in-the-loop training, delivering real-time telemetry at a fraction of commercial tracker costs.

Engineering Evaluation: System Strengths vs. Current Limitations

Core System Strengths

1. **High Affordability:** The ₹885 prototype build cost (scaling to a landed production cost of ~₹2,185 and a commercial retail target of ₹3,000–₹3,500) keeps assistive hardware accessible to low-income households.
2. **Thermal Isolation & Patient Privacy:** Direct battery wiring avoids forehead heat accumulation ($<0.1^\circ\text{C}$ rise), while optical sensors operate without cameras, preserving 100% local patient privacy (zero video data recorded or transmitted).
3. **Cross-Platform Integration:** Runs on low-cost host hardware using a unified Python control service and lightweight web app interface.

Current Limitations & Engineering Mitigations

1. **Eyelid Motor Requirement:** The base hardware requires at least one functional eyelid to execute click actions. For total facial paralysis, Edition 02 will provide pure iris/pupil tracking via low-cost webcams.
2. **Kinematic Muscle Fatigue:** Extended head steering can cause neck fatigue over long sessions. This is mitigated by the 25° dead-zone filter and holding clutch mechanism, allowing users to easily re-center their baseline position.

3. **RF Spectrum Crowding:** In dense hospital wards with heavy 2.4GHz spectrum usage, wireless packet drops can occur. The firmware addresses this by automatically failing over to local Wi-Fi networks or switching to a direct USB-C wired connection.
4. **Regulatory Approval Pathway:** Commercial competitors like Tobii maintain market dominance in critical care units due to extensive clinical history and formal FDA/CE medical device certifications. IRIS is actively progressing through clinical validation pathways at AIIMS Delhi and NIMHANS Bangalore to complete formal regulatory compliance certification.

All hardware schematics, XIAO RISC-V firmware, Python bridge scripts, 3D-printable STL files, and PCB layouts are published under an open-source model at github.com/project-iris/iris and projectiris.dev, providing accessible, hands-free assistive technology worldwide.